

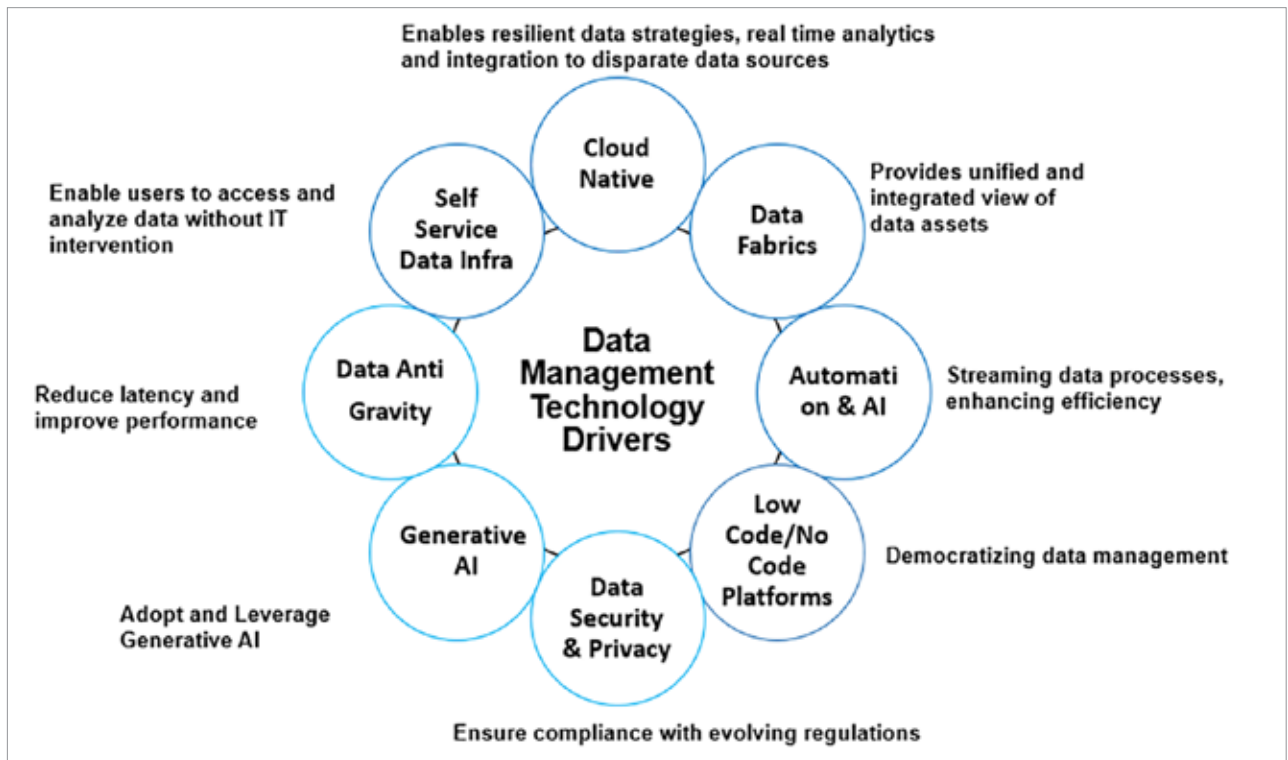


Figure 1: Data management technology drivers

- **Data cleansing** involves the identification and removal of inaccurate, incomplete, or irrelevant data. Duplicate data is removed and incorrect data values are corrected.
- **Data normalisation** helps transform data to a standard format for easy comparison and analysis.
- **Data tokenisation** is a process of breaking the text into individual words or phrases using natural language processing (NLP).

Top open source data processing tools that are widely used include Python, R, Matplotlib, Apache Spark, Apache Hadoop, Apache Kafka and Apache Airflow.

Data quality: This helps to ensure the accuracy, completeness, and consistency of data. Data quality management involves identifying and correcting errors, standardising formats, and maintaining a high level of data integrity.

Some of the best open source data quality tools are Cucumber, Deequ, dbt

Core, MobyDQ, Great Expectations and Soda Core. These tools help automate the data validation process, as well as data cleaning and monitoring.

Data pipeline: This acts like a path to move data from one system to another. ETLs are a popular type of data pipeline. They make it easier for businesses to pull data from multiple sources into a single source. During the process, the data moves through three steps:

- **Extraction:** Pulling data from a database.
- **Transformation:** Manipulating the data with code for formatting and preparing it for analysis.
- **Loading:** Writing the data into the new location.

The leading open source ETL tools include Apache Airflow, Apache Kafka, Data Build Tool, Airbyte, Meltano, Singer and Mage. These tools help to ingest, process, store, transform and visualise data efficiently. They provide scalability, flexibility and active community support.

Data integration: At the enterprise level data may reside in various systems, such as databases, data warehouses, cloud storage, Excel sheets, etc, each with its own structure and semantics. Also, the data may be in different formats, including CSV, XML, JSON, or proprietary formats specific to ERP-centric applications (SAP, Salesforce, etc).

Data integration solutions deal with the aggregation of data from diverse sources and different formats to provide a single view. This eliminates data silos, enhancing cross-functional collaboration and decision-making.

Widely used open source data integration tools are Apache NiFi, Apache Kafka, Apache Flume, Fluentd, Sqoop, Apache Camel, Apache Flink, Logstash, Talend Open Studio and StreamSets. These tools can handle various data integration tasks ranging from simple batch jobs to complex streaming data flows.